

Users > moshika > Desktop > lab > 🐒 monkey.pl

```
1  % State Representation: state(MonkeyPos, MonkeyHasBanana, BoxPos, MonkeyOnBox)
2  % MonkeyPos = floor / onbox
3  % MonkeyHasBanana = has / hasnot
4  % BoxPos & MonkeyPos = door / middle / window
5
6  % Initial State:
7  % Monkey at door, box at window, monkey on floor, does not have banana
8  start :-
9      move(state(door, hasnot, window, floor), Moves),
10     write('Monkey action steps: '), nl,
11     print_moves(Moves).
12
13 % Goal State:
14 move(state(_, has, _, _), []) :- !.
15
16 % Move to box
17 move(state(Pos, hasnot, Box, floor), ['Walk to box' | Moves]) :-
18     Pos \= Box,
19     move(state(Box, hasnot, Box, floor), Moves).
20
21 % Push box to middle if at same place
22 move(state(Pos, hasnot, Pos, floor), ['Push box to middle' | Moves]) :-
23     Pos \= middle,
24     move(state(middle, hasnot, middle, floor), Moves).
25
26 % Climb box
27 move(state(middle, hasnot, middle, floor), ['Climb box' | Moves]) :-
28     move(state(middle, hasnot, middle, onbox), Moves).
29
30 % Monkey grabs banana
31 move(state(middle, hasnot, middle, onbox), ['Grab banana' | Moves]) :-
32     move(state(middle, has, middle, onbox), Moves).
33
34 % Print Steps
35 print_moves([]).
36 print_moves([Step|Rest]) :-
37     write('- '), write(Step), nl,
38     print_moves(Rest).
39
```

```
[moshika@Moshikas-MacBook-Air ~ % cd /Users/moshika/Desktop/lab  
[moshika@Moshikas-MacBook-Air lab % swipl monkey.pl  
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)  
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.  
Please run ?- license. for legal details.
```

```
For online help and background, visit https://www.swi-prolog.org  
For built-in help, use ?- help(Topic). or ?- apropos(Word).
```

```
[?- start.  
Monkey action steps:  
- Walk to box  
- Push box to middle  
- Climb box  
- Grab banana  
true .
```

```
[?- halt.
```